18	**Answer: A**
	$d\sin\theta = n\lambda$ 4 orders are observed and for max angle of 90^o, $d\sin 90^o = (4)(5.50\times10^{-7})$ $d = 2.20\times10^{-6}$ m If 5 orders were to be observed, $d\sin 90^o = (5)(5.50\times10^{-7})$ $d = 2.75\times10^{-6}$ m Larger the value of d results in more orders observed. OR $4 \le \left[\dfrac{d\sin 90^o}{\lambda} = n\right] < 5$ $4\lambda \le d < 5\lambda$ $2.20\times10^{-6} \le d < 2.75\times10^{-6}$

19	**Answer: C**
	$\frac{s}{r}=\frac{\lambda}{b}$ $\frac{s}{20\times3.00\times10^{8}\times365\times24\times60\times60}=\frac{550\times10^{-9}}{3.0}$ s = 3.47 $\times10^{10}$ m

20	**Answer: A**